Math Kangaroo Practice - Grades 3-4

Practice set for 2002 #11
10 practice problems per selected problem.

Problem 1: 2002 #11 - Cracked Clock Face

Partitioning the clock face (sum 1 + 2 + ... + 12 = 78) into pieces with given sum properties

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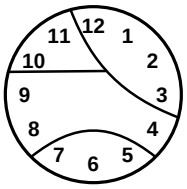
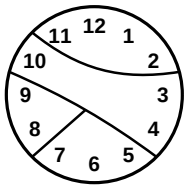
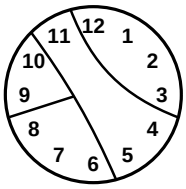
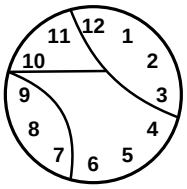
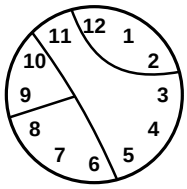
Practice 1.1 The face of a clock is cracked into 4 pieces so that two of the pieces each have a sum of 18, and the other two pieces each have a sum of 21. Which clock face shows this?

A) 

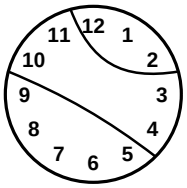
B) 

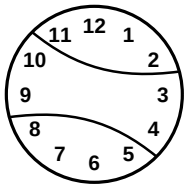
C) 

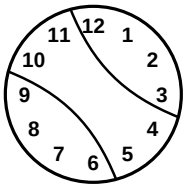
D) 

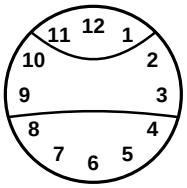
E) 

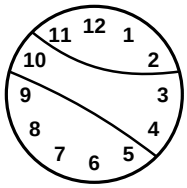
Practice 1.2 The face of a clock is cracked into 3 pieces so that the sums of the numbers on all 3 pieces are equal. Which clock face shows how it was cracked?

A) 

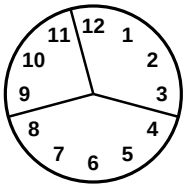
B) 

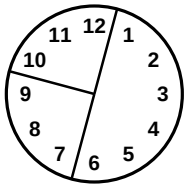
C) 

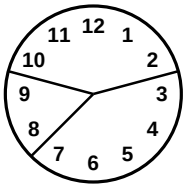
D) 

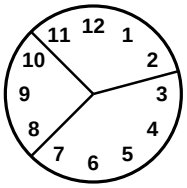
E) 

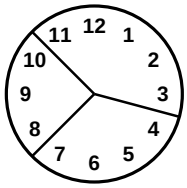
Practice 1.3 The face of a clock is cracked into 3 pieces meeting at the center. The sums of the numbers within the 3 parts are consecutive numbers. Which clock face shows how it was cracked?

A) 

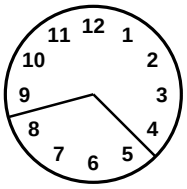
B) 

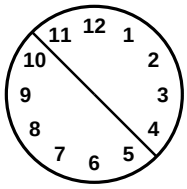
C) 

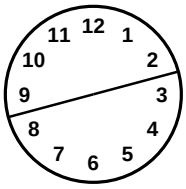
D) 

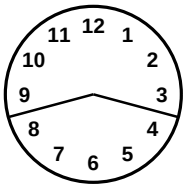
E) 

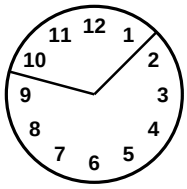
Practice 1.4 A clock face is cracked into 2 pieces by a single straight crack so that the sum of the numbers on the larger piece is exactly **twice** the sum of the numbers on the smaller piece. Which clock face shows this?

A) 

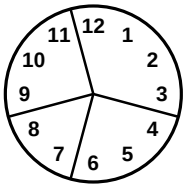
B) 

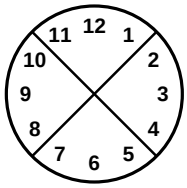
C) 

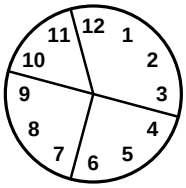
D) 

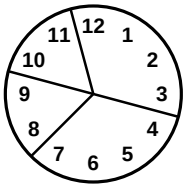
E) 

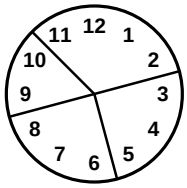
Practice 1.5 The face of a clock is cracked into 4 wedge-shaped pieces meeting at the center. When ordered from smallest to largest, the sums of the pieces increase by 3 each time (15, 18, 21, and 24). Which clock face shows how it was cracked?

A) 

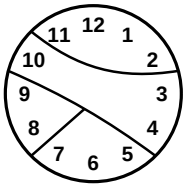
B) 

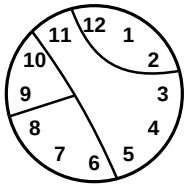
C) 

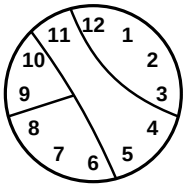
D) 

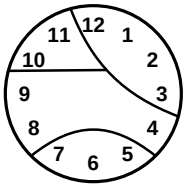
E) 

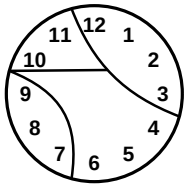
Practice 1.6 The face of a clock is cracked into 4 pieces so that two of the pieces each have a sum of 18, and the other two pieces each have a sum of 21. Which clock face shows this?

A) 

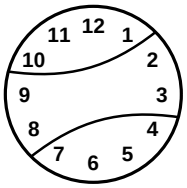
B) 

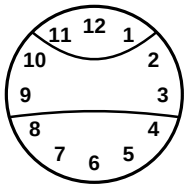
C) 

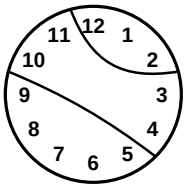
D) 

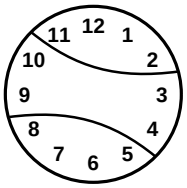
E) 

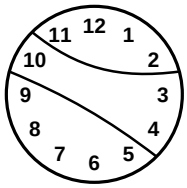
Practice 1.7 The face of a clock is cracked into 3 pieces so that the sums of the numbers on all 3 pieces are equal. Which clock face shows how it was cracked?

A) 

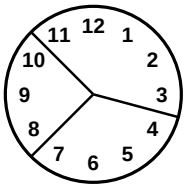
B) 

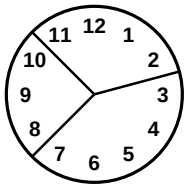
C) 

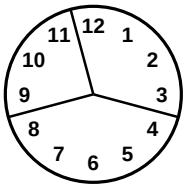
D) 

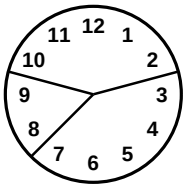
E) 

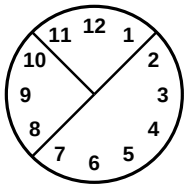
Practice 1.8 The face of a clock is cracked into 3 pieces meeting at the center. The sums of the numbers within the 3 parts are consecutive numbers. Which clock face shows how it was cracked?

A) 

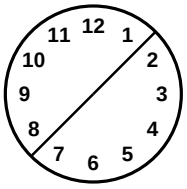
B) 

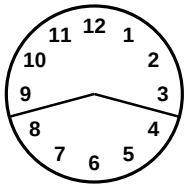
C) 

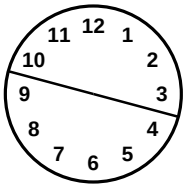
D) 

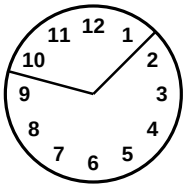
E) 

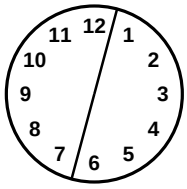
Practice 1.9 A clock face is cracked into 2 pieces by a single crack so that the sum of the numbers on one piece equals the sum of the numbers on the other piece. Which clock face shows this crack?

A) 

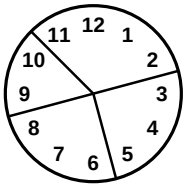
B) 

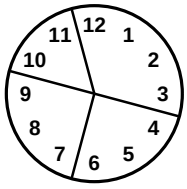
C) 

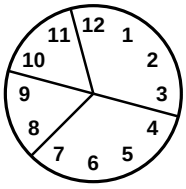
D) 

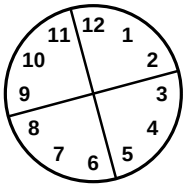
E) 

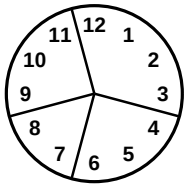
Practice 1.10 The face of a clock is cracked into 4 wedge-shaped pieces meeting at the center. When ordered from smallest to largest, the sums of the pieces increase by 3 each time (15, 18, 21, and 24). Which clock face shows how it was cracked?

A) 

B) 

C) 

D) 

E) 

Answer Key & Explanations

For generated/2026-10-19-21-33-37.pdf (Practice set for 2002 #11)

Solutions & kid-friendly explanations

1.1: A In clock A, the top-right piece has {12, 1, 2, 3} = 18, the bottom piece has {5, 6, 7} = 18, the top-left piece has {10, 11} = 21, and the middle piece has {8, 9, 4} = 21.

1.2: B Since the numbers 1 through 12 add up to 78, each of the 3 equal pieces must have a sum of 78 ÷ 3 = 26. In clock B, the top piece has 11 + 12 + 1 + 2 = 26, the middle piece has 9 + 10 + 3 + 4 = 26, and the bottom piece has 5 + 6 + 7 + 8 = 26.

1.3: D Three consecutive numbers that add up to 78 are 25, 26, and 27 (25 + 26 + 27 = 78). In clock D, the three pieces are {3, 4, 5, 6, 7} = 25, {11, 12, 1, 2} = 26, and {8, 9, 10} = 27.

1.4: A Since the total sum is 78 and one piece has twice the sum of the other, the two sums must be 78 ÷ 3 = 26 and 2 × 26 = 52. In clock A, the pieces are {5, 6, 7, 8} = 26 and {9, 10, 11, 12, 1, 2, 3, 4} = 52.

1.5: C In clock C, the four wedges are {4, 5, 6} = 15, {12, 1, 2, 3} = 18, {10, 11} = 21, and {7, 8, 9} = 24. Sorted in order, the sums are 15, 18, 21, and 24.

1.6: D In clock D, the top-right piece has {12, 1, 2, 3} = 18, the bottom piece has {5, 6, 7} = 18, the top-left piece has {10, 11} = 21, and the middle piece has {8, 9, 4} = 21.

1.7: D Since the numbers 1 through 12 add up to 78, each of the 3 equal pieces must have a sum of 78 ÷ 3 = 26. In clock D, the top piece has 11 + 12 + 1 + 2 = 26, the middle piece has 9 + 10 + 3 + 4 = 26, and the bottom piece has 5 + 6 + 7 + 8 = 26.

1.8: B Three consecutive numbers that add up to 78 are 25, 26, and 27 (25 + 26 + 27 = 78). In clock B, the three pieces are {3, 4, 5, 6, 7} = 25, {11, 12, 1, 2} = 26, and {8, 9, 10} = 27.

1.9: C Half of the total clock sum is 78 ÷ 2 = 39. In clock C, the crack goes between 9 and 10 on the left and between 3 and 4 on the right, giving {10, 11, 12, 1, 2, 3} = 39 and {4, 5, 6, 7, 8, 9} = 39.

1.10: B In clock B, the four wedges are {4, 5, 6} = 15, {12, 1, 2, 3} = 18, {10, 11} = 21, and {7, 8, 9} = 24. Sorted in order, the sums are 15, 18, 21, and 24.